We can write x_1, x_2 in terms of y_1, y_2 as

$$x_1 = \sqrt{y_1 y_2} \qquad x_2 = \sqrt{\frac{y_2}{y_1}}$$

Then we can re-write the joint probability distribution as

$$f\left(\sqrt{y_1 y_2}, \sqrt{\frac{y_2}{y_1}}\right) = \begin{cases} 4\sqrt{y_1 y_2 \frac{y_2}{y_1}} & 0 < \sqrt{y_1 y_2} < 1, \; 0 < \sqrt{\frac{y_2}{y_1}} < 1 \\ 0 & \text{otherwise} \end{cases}$$

Figuring out the bounds:

since all positive

$$0 < \sqrt{y_1 y_2} < 1 \Rightarrow 0 < y_1 y_2 < 1$$

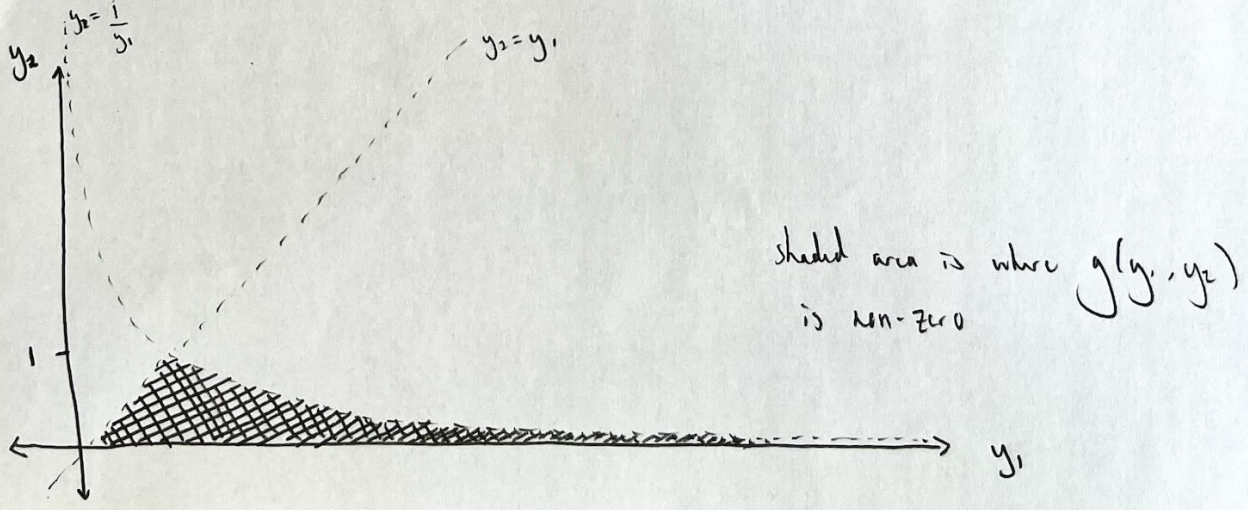
$$0 < \sqrt{\frac{y_2}{y_1}} < 1 \Rightarrow 0 < \frac{y_2}{y_1} < 1$$

$$\Rightarrow 0 < y_1 < \frac{1}{y_2} \quad \text{and} \quad 0 < y_2 < y_1$$

$$\Rightarrow 0 < y_2 < y_1 < \frac{1}{y_2}$$

Therefore

$$g(y_1, y_2) = \begin{cases} 4y_2 & 0 < y_2 < y_1 < \frac{1}{y_2} \\ 0 & \text{otherwise} \end{cases}$$



shaded area is where $g(y_1, y_2)$ is non-zero